

Solution Requirements

Date	27 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The current version does not require user authentication. Future versions may support secure login for personalized learning history.	Low
2	Authorization Levels	The current system provides a single user role. Future versions may include Admin, Faculty, and Student roles.	Low
3	External Interfaces	The system integrates with Google Gemini API, TensorFlow BiLSTM, Hugging Face BERT, and Streamlit for emotion analysis and AI-generated learning support.	High

4	Emotion Detection	The application accepts student text, preprocesses it, predicts emotions using BiLSTM and BERT, and generates personalized learning guidance.	High
5	Reporting	The system generates emotion analysis reports, dashboards, charts, and downloadable prediction history.	Medium
6	Business Rules	Users must enter valid text. The system preprocesses the input, predicts emotions, and generates empathetic learning responses based on detected emotions.	High
7	Compliance to Laws or Regulations	The application stores only essential learning data and follows data privacy principles while protecting user information.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>[Describe response times, processing speed, etc.]</i>	<i>(e.g., Page loads in < 2 seconds)</i>
2	Scalability	<i>[Describe how the system handles growth in users or data]</i>	<i>(e.g., Supports up to 10,000 concurrent users)</i>
3	Security & Data Privacy	<i>[Detail data protection, encryption, and vulnerability management]</i>	<i>(e.g., End-to-end AES-256 encryption)</i>
4	Reliability & Availability	<i>[Specify uptime requirements and fault tolerance]</i>	<i>(e.g., 99.9% uptime per month)</i>
5	Usability & Accessibility	<i>[Describe user experience standards and accessibility compliance]</i>	<i>(e.g., Complies with WCAG 2.1 AA standards)</i>
6	Other	<i>[Add any additional non-functional requirements (e.g., Maintainability, Portability)]</i>	